Fixed-Ro equal-area theory on the equatorial β plane, for the SS09 shallow-water model

pengcheng-ss09 suite, Tier 0 (A0 memo)

2026-07-16

Beta-plane translations of the closed forms in the fixed-Ro manuscript (`papers/fixed-ro-eq-area_ms-draft-v1.pdf`, "ms" below), derived against this model's exact equations. Every result carries a basis tag; "pending run" tags get upgraded as the suite's runs confirm them. Section 8 lists the falsifying checks and their status, and the run record follows it.

1 Setup and notation

Steady state, hemispherically symmetric forcing ($y_0 = 0$), northern hemisphere ($y > 0$). The model's steady equations (`SCIENCE.md` section 2), with $T = \theta/1.6$ and the v equation's factor of two multiplying only the time tendency (so it drops out of every steady balance):

$$v\left(\beta y - \frac{\partial u}{\partial y}\right) = \mathcal{H}(\theta_E - \theta)\, u\frac{\partial v}{\partial y} + \epsilon_u u + v_d\, \mathcal{H}(u)\,\mathrm{sgn}(y)\frac{\partial u}{\partial y}, \tag{1}$$

$$\beta y\, u = -\frac{gH}{T_0}\frac{\partial T}{\partial y} + k_v\frac{\partial^2 v}{\partial y^2}, \tag{2}$$

$$\frac{\delta\Delta_z}{H}\frac{\partial v}{\partial y} = \frac{\theta_E - \theta}{\tau}. \tag{3}$$

Forcing (SS09 parabolic profile, the theory-matched choice): $\theta_E = \theta_{00} - \Delta_y(y/y_1)^2$ for $|y| < y_1$. In temperature units,

$$T_E(y) = T_E(0) - b\, y^2, \qquad b \equiv \frac{\Delta_T}{y_1^2}, \qquad \Delta_T \equiv \frac{\Delta_y}{1.6}, \tag{4}$$

with $T_0 = 300$ K. The small-angle limit of the ms is exact here: the β plane has no $\cos\varphi$ factors, so no truncation enters any of the following.

2 The one-line mapping

Every ms result carries over with the substitutions

$$\mathrm{Ro_{th}} \;\to\; R_\beta \equiv \frac{4gH\Delta_T}{T_0\beta^2 y_1^4}, \qquad \Delta_h\theta_0 \;\to\; \Delta_T, \qquad a\varphi \;\to\; y, \qquad \varphi_{\mathrm{Ro}} \;\to\; Y_{\mathrm{Ro}}/y_1. \tag{5}$$

[derived] R_β follows from the sphere definition $\mathrm{Ro_{th}} = gH\Delta_h/(\Omega a)^2$ with $\Omega = \beta a/2$ and $\Delta_h = \Delta_T a^2/(T_0 y_1^2)$ (matching the forcing curvature at the equator); the planetary radius cancels, as it must on a β plane. At this repo's defaults ($g = 9.81\ \mathrm{m\,s^{-2}}$, $H = 16$ km, $\Delta_y = 50$ K so $\Delta_T = 31.25$ K, $T_0 = 300$ K, $\beta = 2\times10^{-11}\ \mathrm{m^{-1}\,s^{-1}}$, $y_1 = 9439$ km):

$$\boxed{R_\beta = 0.0206.} \tag{6}$$

3 Wind and temperature

[derived: pending run confirmation] Uniform $\mathrm{Ro} = (\partial_y u)/(\beta y)$ integrates to $u_{\mathrm{Ro}}(y) = \mathrm{Ro}\,\beta y^2/2$ (ascent at $y = 0$). Gradient balance (neglecting k_v) gives $\partial_y T = -(T_0/gH)\,\mathrm{Ro}\,\beta^2 y^3/2$, so

$$T_{\mathrm{Ro}}(y) = T_{\mathrm{Ro}}(0) - c\,y^4, \qquad c \equiv \frac{\mathrm{Ro}\,T_0\beta^2}{8gH}. \tag{7}$$

4 Equal-area edge and equatorial temperature

[derived: confirmed by runs 1a–1b at the equilibrated day-6000 record: depression within 3%, edge −7 to −8%; at profile level the inner half of the cell agrees at the 10% level and the terminus misfit is ∼33–36% of the depression; run-record addenda] With $D(y) = T_{\mathrm{Ro}}(y) - T_E(y)$, imposing $D(Y) = 0$ and $\int_0^Y D\,\mathrm{d}y = 0$ (measure $\mathrm{d}y$; the Newtonian relaxation has uniform τ) gives $\frac{2}{3}bY^2 = \frac{4}{5}cY^4$, i.e. $Y^2 = \frac{5}{6}\,b/c$, which in Section-2 variables is

$$Y_{\mathrm{Ro}} = y_1\left(\frac{5R_\beta}{3\mathrm{Ro}}\right)^{1/2} \qquad \text{[ms Eq. 13 analog]}, \tag{8}$$

$$T_E(0) - T_{\mathrm{Ro}}(0) = \frac{5}{18}\frac{R_\beta\,\Delta_T}{\mathrm{Ro}} \qquad \text{[ms Eq. 15 analog]}. \tag{9}$$

Defaults, $\mathrm{Ro} = 1$: $Y_{\mathrm{Ro}} = 1.75 \times 10^6$ m and equatorial depression 0.18 K (in T; multiply by 1.6 for θ units). At $\mathrm{Ro} = 0.15$ the depression is 1.2 K (T) / 1.9 K (θ).

Consistency note: the analogous derivation for the $\sin^2$ profile needs the numerical equal-area solver (puffins `equal_area_bouss` with Ro-scaled `del_h_over_ro`, verified in puffins PR #46). Near the equator $\sin^2\big(\pi y/(2y_1)\big) = \big(\pi y/(2y_1)\big)^2 + O(y^4)$, so $\sin^2$ behaves like a parabola with curvature enhanced by $(\pi/2)^2 = 2.47$; Section 7's postdiction check uses exactly this factor.

5 Meridional wind, heat flux, momentum flux

[derived: confirmed by runs 1a–1b at the equilibrated day-6000 record: $v_{\max}$ within 2–6% and its location within 4–5%; at profile level the inner flank agrees at the 3–7% level and the terminus flank misfit is ∼43–45% of $v_{\max}$; run-record addenda] The steady thermodynamic balance (3) integrates to the model's v profile directly, using $\theta_E - \theta = 1.6\,(T_E - T) = -1.6\,D$:

$$v(y) = -\frac{1.6H}{\delta\Delta_z\tau}\int_0^y D\,\mathrm{d}y' = \frac{1.6H}{\delta\Delta_z\tau}\,\frac{5}{18}\,\frac{R_\beta\Delta_T}{\mathrm{Ro}}\,Y_{\mathrm{Ro}}\left(x - 2x^3 + x^5\right), \qquad x \equiv \frac{y}{Y_{\mathrm{Ro}}}. \tag{10}$$

The bracket is exactly the ms Eq. (16) shape; it peaks at $x = 5^{-1/2} = 0.447$ with value 0.286, and $v_{\max} \propto \mathrm{Ro}^{-3/2}$ (ms Eq. 20 analog). Defaults, $\mathrm{Ro} = 1$: $v_{\max} = 3.0 \times 10^{-3}\ \mathrm{m\,s^{-1}}$ at $y = 0.78 \times 10^6$ m.

Column fluxes, using the two-layer reading (upper slab of depth δ with velocity $+v$ and potential temperature θ_{top}, return slab of depth δ with $-v$ and θ_{bot}, $\theta_{\mathrm{top}} - \theta_{\mathrm{bot}} = \Delta_z$; return-flow u negligible):

$$\text{heat flux:} \quad \int v\,\theta\,\mathrm{d}z = \delta\,\Delta_z\,v(y) \qquad \text{[ms Eq. 16]}, \tag{11}$$

$$\text{momentum flux:} \quad \int v\,u\,\mathrm{d}z = \delta\,v(y)\,u_{\mathrm{Ro}}(y) \qquad \text{[ms Eq. 17]}, \tag{12}$$

$$\text{vertical velocity:} \quad w = \delta\frac{\partial v}{\partial y}, \qquad w_{\max} \propto \mathrm{Ro}^{-1} \qquad \text{[ms sec. 3f]}. \tag{13}$$

The ms's counterintuitive momentum-flux result carries over: expanding $\delta v u_{\mathrm{Ro}}$, the leading term in y is Ro-independent (the Ro^{-1} strengthening of v cancels the Ro weakening of u in the product) and $\partial(\mathrm{flux})/\partial\mathrm{Ro} < 0$ strictly inside the cell, exactly as verified for the sphere forms in puffins `test_eq_area.py`.

6 Diagnosed quantities: conventions for the suite

- **Full-profile scorecard (the confirmation standard for every suite run):** the simulated u, T, and v profiles are compared against the zero-free-parameter theory profiles at the diagnosed (regression) cell-mean Ro, reporting $\max|\Delta|$ and its location both over the cell $[0, Y_{\mathrm{Ro}}]$ and over its inner half, each scaled by the field's theory amplitude (cell-edge u, the equatorial depression, and $v_{\max}$ respectively). The temperature comparison is visualized as the equal-area lobes $D(y) = T - T_E$, which removes the ∼200 K shared background. Implemented in `scripts/fixed_ro_scorecard.py` (one four-panel figure per run; fields are means over the final 30 days). Scalar anchors (depression, $v_{\max}$, edge) are by-products of the same tool, not the confirmation standard: runs 1a–1b passed the scalars at the 1–8% level while the profile misfit at the terminus is ∼35–45% of scale (see the profile addendum in the run record).

- **Run duration (budget-grade):** the steady-state detector's KE and T-variance metrics are nearly blind to the slow jet-position mode: in runs 1a–1b the detector fired at day ∼960 with the jet latitude still moving ∼0.7% per 60 d (measured over the final 120 d). Scorecard-grade runs therefore use a fixed length of at least five times the slowest damping timescale, the drag time $1/\epsilon_u$ (1157 d at the default $\epsilon_u = 10^{-8}$ s^{-1}, so 6000 d), with no early stop. The scorecard enforces this after the fact: it fits the jet-latitude trend over the final 120 d and warns unless the implied remaining drift (trend $\times\ 1/\epsilon_u$, the exponential-tail estimate) is below 0.2%. The detector remains fine for smokes and exploratory runs.

- **Local** $\mathrm{Ro}(y)$**:** $(\partial_y u)/(\beta y)$, the model's existing diagnostic (NaN within $|y| < \Delta y_{\mathrm{grid}}$ of the equator).

- **Cell-mean** Ro**, two definitions, both reported** (paralleling Hill et al. 2025): (i) the area mean of local Ro over $[y_{\text{Ro-max}}, Y_{\mathrm{edge}}]$; (ii) the best-fit Ro regressing $u(y)$ against $\beta y^2/2$ over the same interval. The near-ascent region is excluded because $\mathrm{Ro} \to 0$ at the equator whenever $v_d > 0$ (Zhang et al. 2025, $u \sim y^3$).

- **Cell edge, three definitions, all reported:** the v-threshold edge ($|v|$ below 10% of its extremum, existing diagnostic; see run 1a for its failure mode at small v_d), the jet latitude (existing spline diagnostic), and the θ-merge latitude (where θ rejoins θ_E within a tolerance, searched poleward of the *jet*; see run 1a). The theory says they coincide; their splitting is itself a result.

- **Momentum-budget shares:** each steady term of (1) evaluated from output; the drag share $\epsilon_u u$ and the vertical-advection share quantify the accuracy of the EMFD–Coriolis balance underlying Zhang et al. (2025) Eq. 24, $\mathrm{Ro} = v/(v + v_d)$.

7 Emergent cell-mean Ro closure

[**derived: pending run confirmation**] Combining Zhang's Eq. 24 at the cell-mean level with Section 5's v magnitude (the cell mean of the bracket over $[0, 1]$ is $1/6$):

$$\bar{\mathrm{Ro}} = \frac{\bar{v}}{\bar{v} + v_d}, \qquad \bar{v}(\bar{\mathrm{Ro}}) = \frac{1.6H}{\delta\Delta_z\tau}\,\frac{5}{18}\,\frac{R_\beta\Delta_T}{\bar{\mathrm{Ro}}}\,\frac{Y_{\mathrm{Ro}}(\bar{\mathrm{Ro}})}{6}, \tag{14}$$

a transcendental equation for $\bar{\mathrm{Ro}}(v_d)$ with no simulation input. This is the suite's analysis G; it extends Zhang et al. (2025) section 4 (their max-Ro scalings $v_d^{-2/5}$, $v_d^{-1/4}$) to the cell-mean variable the ms uses.

Postdiction check [**computed**]: Section 5's v with the $\sin^2$ profile's larger near-equator curvature ($(\pi/2)^2 = 2.47$ in b, entering v through b^2Y/c as a factor ~ 9.6) predicts $v_{\max} \sim 0.03$ m s^{-1} for the published defaults, matching Zhang et al. (2025) Figs. 2–3 magnitudes within $\sim$1.5×.

8 Falsifying checks and status

Every run in the checks below also gets the Section 6 full-profile scorecard on u, T, and v; the tier-specific items are the additional, targeted comparisons.

1. **Full $u/T/v$ profiles, edge, and equatorial depression vs theory at diagnosed Ro (the most-falsifying single check):** SS09-parabolic forcing, $v_d = 0$. Framing correction from the first run: with the default vertical momentum advection on, $v_d = 0$ is *not* the $\mathrm{Ro} = 1$ limit (Zhang et al. 2025: $u \sim \beta y^2/3$ near the ascent, and drag lowers it further); the theory must be evaluated at the run's diagnosed cell-mean Ro. The $\mathrm{Ro} \to 1$ anchor requires `--no-vert-advec-u` *and* reduced drag (run 1b). Status: runs 1a–1b below, scalar level and profile level (addendum).

2. **Edge collapse:** $Y_{\mathrm{edge}}(\mathrm{Ro}/R_\beta)^{1/2}/y_1 = (5/3)^{1/2}$ flat across the v_d, Δ_y, β, and radius ladders. Status: pending Tiers 1–2.

3. **Radius-ladder null:** dimensional Y_{edge} invariant under $(\beta \to \beta/k,\ y_1 \to ky_1,\ L \to kL)$. Status: pending Tier 2.

4. **v-profile shape and $\mathrm{Ro}^{-3/2}$ magnitude scaling** (Section 5). Status: shape checked at profile level in runs 1a–1b (addendum); the $\mathrm{Ro}^{-3/2}$ scaling pending Tier 1.

5. **Momentum flux increases as Ro decreases** (Section 5). Status: pending Tier 1.

6. **Emergent $\bar{\mathrm{Ro}}(v_d)$** (Section 7). Status: pending Tier 1 + analysis G.

Run record

Each confirming or refuting run gets an entry: configuration, output path, measured vs predicted.

2026-07-16, check 1a: $v_d = 0$, default physics (vertical advection on)

Config: SS09 parabolic forcing, $v_d = 0$, $n_y = 801$, $\Delta t = 30$ s, numba backend, steady-state stop (window 30 d, threshold 10^{-4}), converged day 959. Output: `model_output/fixed_ro_suite/`

`tier0_amc_edge/amc_ss09_ny801_dt30_vd0.nc`. Steadiness caveat [**computed**]: the jet was still drifting +0.4% per 60 d at stop (drag-timescale tail), so numbers carry roughly a 2% allowance.

Diagnosed: cell-mean Ro 0.469 (area mean) / 0.470 (u regression); local Ro(y) is *not* uniform: ∼0.68 at the ascent (the vertical-advection 2/3 of Zhang et al. 2025, drag-reduced), declining through ∼0.6 mid-cell to ∼0.2 approaching the jet, i.e. the ms's linear-Ro motivation in miniature.

Theory at diagnosed Ro = 0.470 vs run [**computed**]:

quantity	predicted	measured	rel. gap
equatorial depression $T_E(0) - T(0)$	0.381 K	0.379 K	0.5%
$v_{\max}$	9.28×10^{-3} m/s	9.39×10^{-3} m/s	1.2%
y of $v_{\max}$ ($0.447\,Y_{\mathrm{Ro}}$)	1.14×10^6 m	1.18×10^6 m	3.5%
cell edge Y_{Ro} (vs jet latitude)	2.55×10^6 m	2.36×10^6 m	−7.4%

Sections 4–5 upgraded to [**confirmed at the 1–8% level, single run**] for these observables at this Ro. Additional findings: the u plateau outside the cell is the parabolic forcing's RCE gradient wind,

$$u_{\mathrm{RCE}} = \frac{2gH\Delta_T}{T_0\beta y_1^2} = 18.35 \text{ m/s}, \tag{15}$$

constant out to y_1 (verified to 3 digits); the v-threshold edge diagnostic is unusable at small v_d because the drag circulation $v = \epsilon_u u_{\mathrm{RCE}}/(\beta y) \sim 2 \times 10^{-3}$ m/s exceeds 10% of $v_{\max}$ out to $\sim 9.3 \times 10^6$ m; the θ-merge diagnostic must search poleward of the jet, not of the v max, to avoid the equal-area lobe crossing.

2026-07-16, check 1b: $v_d = 0$, vertical advection off

Config: as 1a plus `--no-vert-advec-u`; converged day 971. Output: `.../tier0_amc_edge/amc_ss09_ny801_dt30_vd0_novert.nc`.

Removing vertical advection left the cell-mean Ro essentially unchanged (fit 0.473 vs 0.470; area mean 0.536 vs 0.469), because **the Rayleigh drag, at its default $\epsilon_u = 10^{-8}$ s^{-1}, is the leading-order non-AMC agent for this forcing** [**computed**]: the steady zonal momentum balance over the cell interior is

$$v\left(\beta y - \frac{\partial u}{\partial y}\right) = \epsilon_u u \quad \text{to within} \sim\text{5–10\%} \tag{16}$$

(median ratio 1.05 over $0.3 \times 10^6 < y < 2.2 \times 10^6$ m). The parabolic forcing's weak circulation ($v_{\max} \sim 9 \times 10^{-3}$ m/s) makes $(1 - \mathrm{Ro}) \sim \epsilon_u u/(v\beta y)$ order one. Local Ro(y) declines monotonically from ∼1 at the ascent to ∼0.2 at the jet.

Theory at diagnosed Ro = 0.473 vs run [**computed**]: depression 0.378 predicted vs 0.358 K measured (5.3%); $v_{\max}$ 9.18×10^{-3} vs 8.81×10^{-3} m/s (4.2%); edge 2.54×10^6 vs jet 2.36×10^6 m (−7.1%).

Consequences for the suite: (i) the Ro → 1 anchor needs reduced ϵ_u, so the Tier-3 drag ladder is load-bearing rather than optional; (ii) at the $\sin^2$ forcing the circulation is ∼3× stronger, so the drag share is ∼3× smaller but still non-negligible at $v_d = 0$ (consistent with Zhang et al. 2025 Fig. 4's max Ro ∼ 0.9 there); (iii) the steady-state detector (window 30 d, 10^{-4}) fires while the jet still drifts ∼0.4% per 60 d on the drag timescale, so budget-grade runs need a stricter threshold or a fixed post-detection extension.

2026-07-16, profile-level addendum to runs 1a–1b

The tables above are endpoint checks (the equatorial depression and $v_{\max}$ are, respectively, the $y = 0$ anchor of T and a single point of v). The suite standard is now the full-profile scorecard (Section 6), applied retroactively to both runs: final-30-day means, theory at the fitted cell-mean Ro of 0.479 (1a) and 0.473 (1b). Under this standardized window 1b's fit reproduces the value quoted above exactly, while 1a's shifts from 0.470 to 0.479, within run 1a's stated ∼2% drift allowance. Profile results [**computed**]:

run	field	max $\|\Delta\|$ (cell)	at y [10^6 m]	% of scale	inner half
1a	u	10.4 m/s	2.52	34	2.3 m/s (7.5%)
1a	T	0.133 K	2.01	36	0.038 K (10%)
1a	v	4.0 mm/s	2.44	45	0.49 mm/s (5.5%)
1b	u	10.2 m/s	2.52	33	3.2 m/s (10%)
1b	T	0.152 K	2.01	40	0.042 K (11%)
1b	v	3.9 mm/s	2.44	43	0.48 mm/s (5.2%)

Scales: cell-edge $u_{\mathrm{Ro}}(Y_{\mathrm{Ro}})$, the equatorial depression, and $v_{\max}$. Reading: the inner half of the cell agrees at the 5–11% level in all three fields, and the misfit concentrates at the terminus in all three, consistent with the diagnosed local $\mathrm{Ro}(y)$ declining to ∼0.2 at the jet: a single cell-mean Ro is furthest from the local value exactly there. The simulated subtropical warm lobe of $D(y)$ peaks at 0.17 K (1a) against the predicted 0.30 K and is smeared poleward across the edge (D stays ∼+0.015 K out to at least 4.5×10^6 m), suggesting the sharp-edge equal-area closure, not the quartic interior shape, is the limiting idealization. The u misfit sits at the theory's terminus discontinuity (the jump to the RCE plateau). The ms's linear-Ro variant targets exactly this regime and is testable against these same stored profiles (suite extension, not yet scheduled).

2026-07-17, drift fix: runs 1a–1b extended to day 6000

The detector-stop states above were drift-contaminated: at the day ∼960 stop the jet latitude was still moving −0.69% per 60 d (fit over the final 120 d; equatorward), which the exponential-tail estimate turns into ∼13% remaining drift. (This measurement supersedes the "+0.4% per 60 d" figure quoted in the 1a entry, which came from a different ad-hoc check.) Both runs were therefore restarted from their day-959/971 checkpoints and integrated to day 6000, about five drag timescales, with a fixed length and no early stop, per the Section 6 duration policy. The `..._day6000.nc` outputs are now the citable record for checks 1a–1b; both pass the drift gate with a flat jet trend (+0.0000% per 60 d) and the fitted Ro stable to $< 0.001\%$ between consecutive 30-d windows.

What the contamination had been hiding [**computed**]: the jet settled equatorward (1a: 2.36 → 2.33; 1b: 2.38 → 2.33×10^6 m), so the edge gap vs Y_{Ro} *widens* to −7.9% (1a) / −7.2% (1b); 1b's fitted Ro rises 0.473 → 0.485 (+2.5%, the largest single shift); the depression gaps become +2.9% (1a: measured 0.385 vs predicted 0.374 K) and −0.8% (1b: 0.366 vs 0.369 K); and $v_{\max}$ gaps become 5.6% (1a: 9.58 vs 9.04 mm/s) and 2.0% (1b: 9.03 vs 8.85 mm/s). Equilibrated profile metrics, superseding the table above:

run	field	max $\lvert\Delta\rvert$ (cell)	at y [10^6 m]	% of scale	inner half
1a	u	10.0 m/s	2.52	33	1.9 m/s (6.1%)
1a	T	0.122 K	2.01	33	0.035 K (9.5%)
1a	v	3.9 mm/s	2.44	43	0.64 mm/s (7.1%)
1b	u	9.6 m/s	2.48	32	2.5 m/s (8.1%)
1b	T	0.133 K	1.97	36	0.039 K (10.5%)
1b	v	3.9 mm/s	2.40	45	0.26 mm/s (2.9%)

The profile-level story is unchanged in character: inner half at the 3–11% level, terminus at 32–45% of scale, in all three fields of both runs. One diagnostics finding: during the slow tail, 1a's local-Ro maximum migrated from the ascent ($y \approx 0.04 \times 10^6$ m) to mid-cell (1.14×10^6 m), which collapses the area-mean definition's averaging interval $[y_{\text{Ro-max}}, y_{\text{jet}}]$ and drops its value to 0.355 while the regression value barely moves ($0.479 \to 0.478$). The regression definition is the robust one and stays primary; report the area mean with its interval.